

1. Install Hadoop and Implement the following file management tasks in Hadoop:

Adding Files and directories

Retrieve files

Deleting files and directories

Hint: A typical hadoop workflow creates data files (such as log files) elsewhere and copies them into HDFS using one of the above command line utilities.

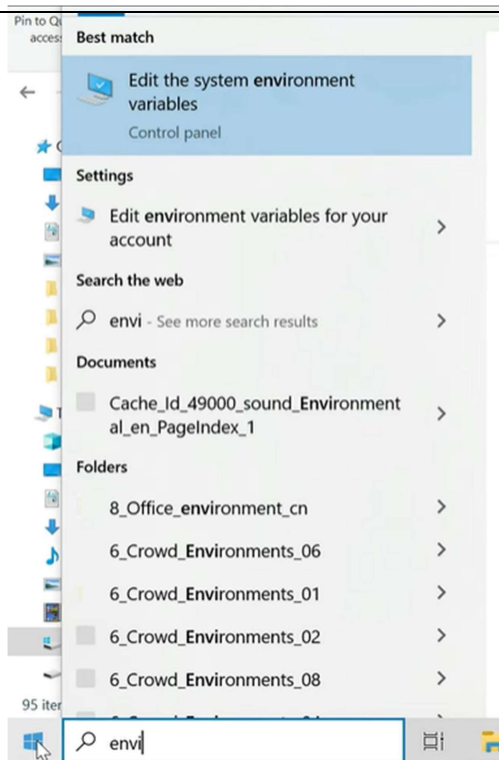
➤ Uninstalling JDK if installed.

1. Hadoop supports on java version 8, so existing version of java need to be deleted.
2. Control panel -> programs -> programs and features -> Right click on Java™ SE development kit -> click on uninstall.

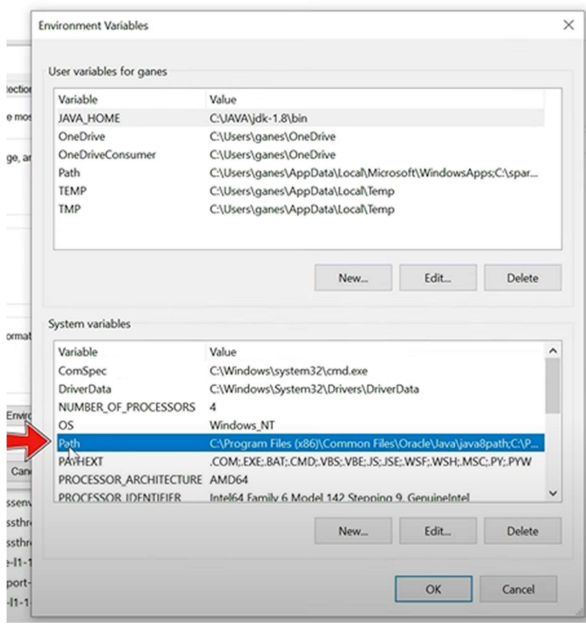
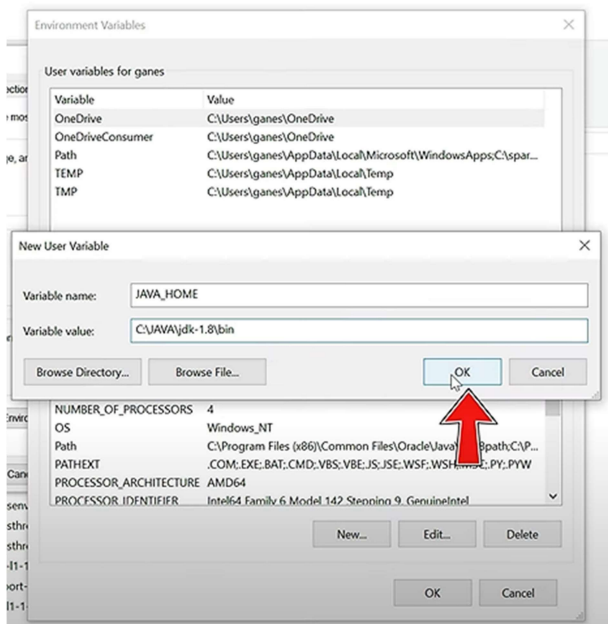
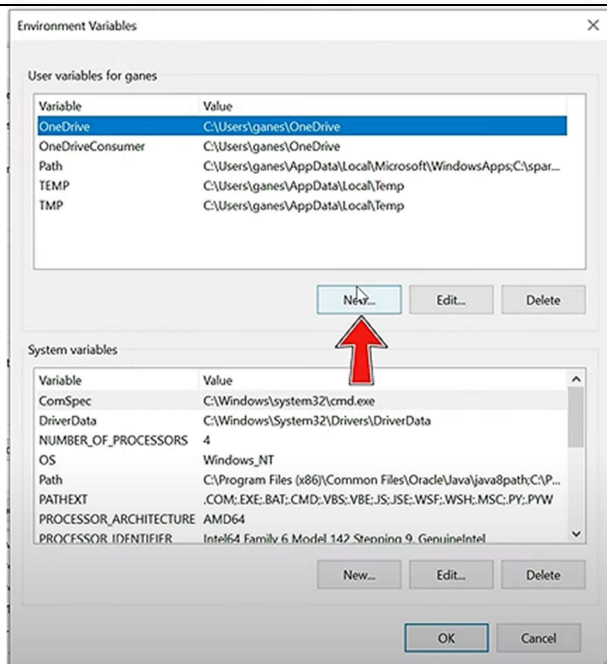
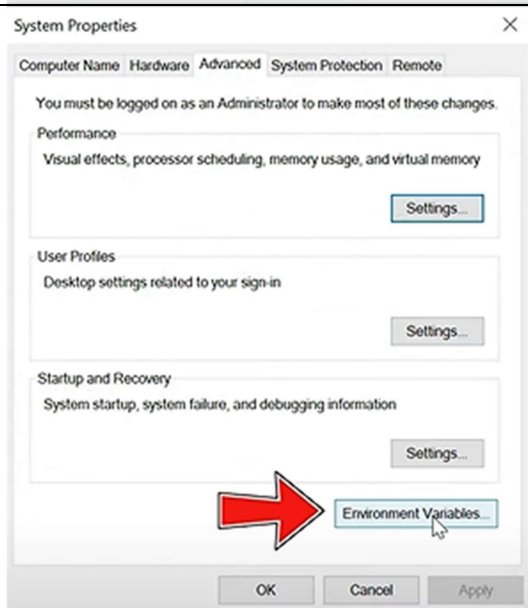
➤ Downloading and Installing Java and Hadoop

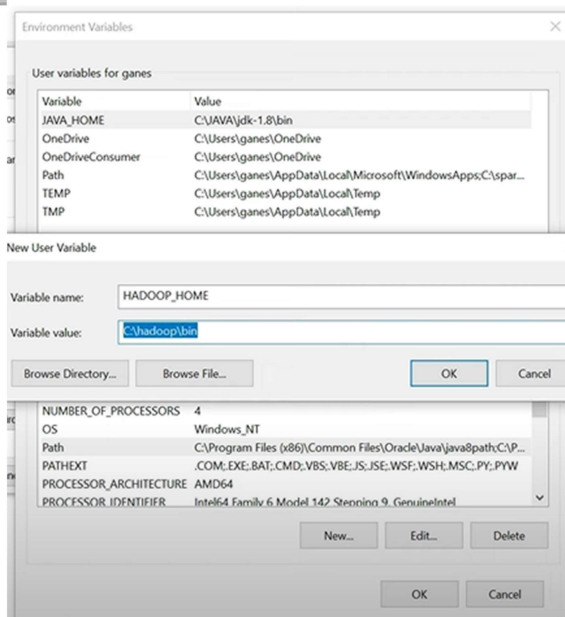
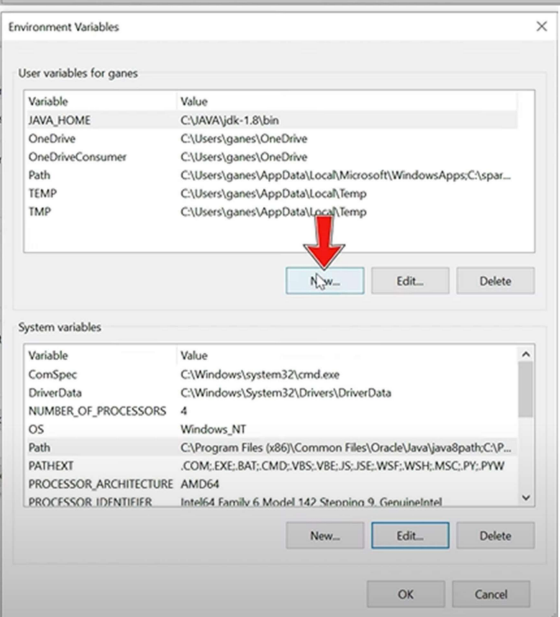
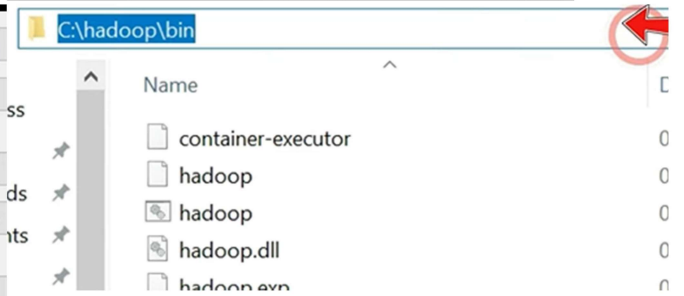
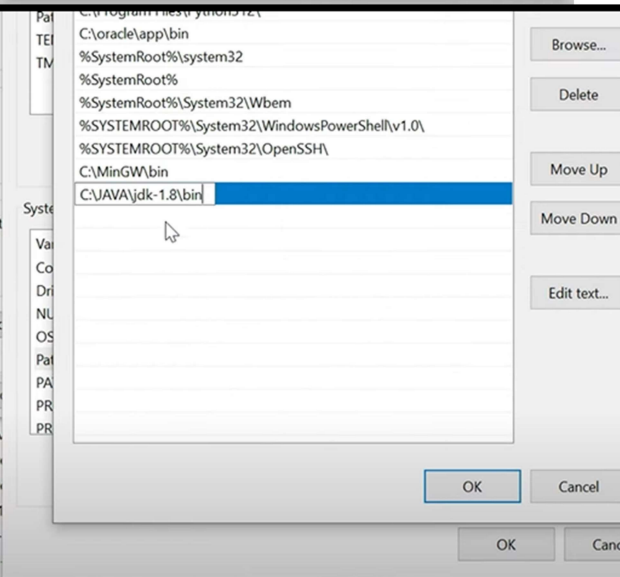
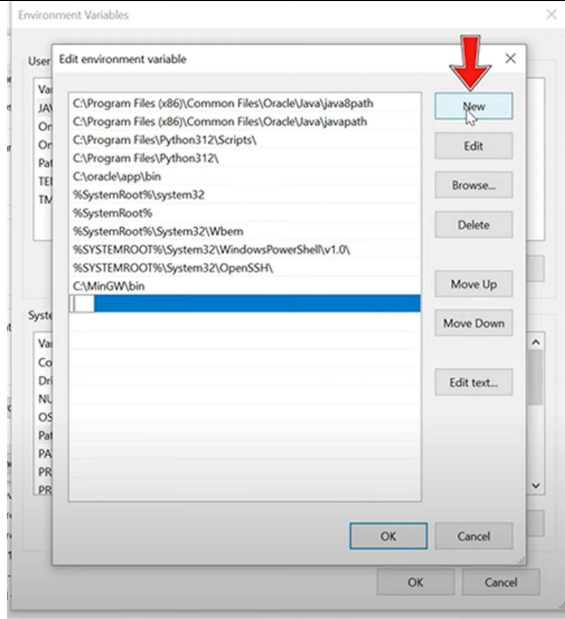
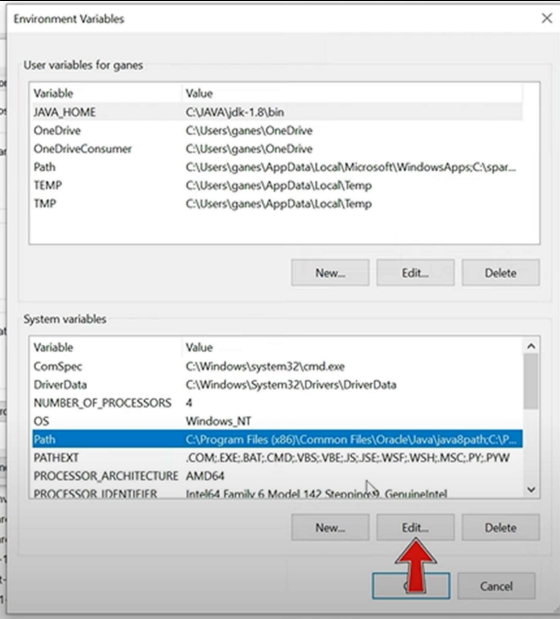
1. In C drive, create a folder named "JAVA"
2. Open a file "Win RAR" and install it.
3. Extract "Technical windows" zip file, by right clicking the zip file and choose extract here.
4. Install JDK-8 by double clicking the icon named "jdk-8" which was in folder named "Technical windows". During installation process, change the destination folder from "C:\Program Files\Java\jre-1.8" to "C:\JAVA". Delete a "JAVA" folder that was created before in C drive. And also, in C drive, within program files, select folder named "Java", and delete it.
5. From the extracted folder "Technical windows" with in that, select "setup files", copy it, and then paste it in C drive. Extract the "setup files" in the same C drive.
6. Open C->JAVA->jdk-1.8->bin, copy the path.
7. Search for "Environmental Variables" options by typing in search bar. Look into an image given.

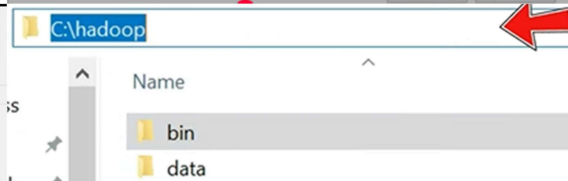
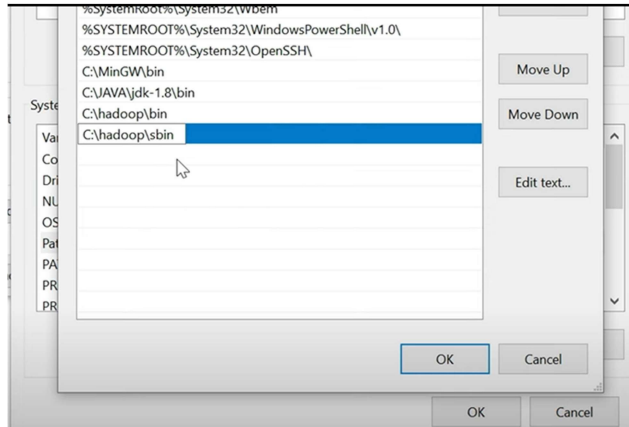
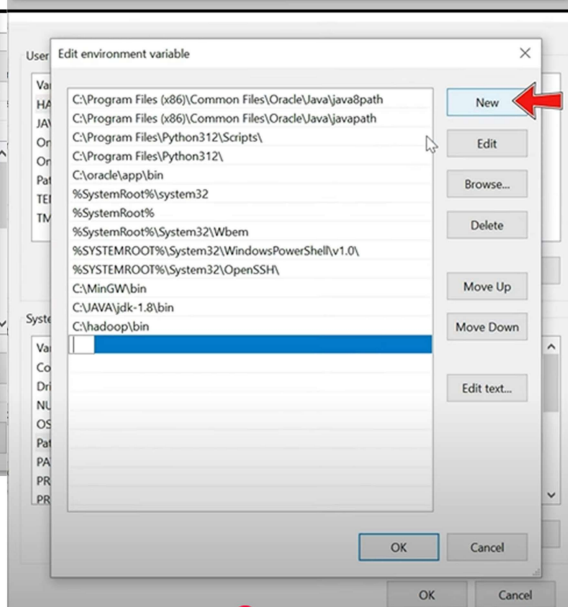
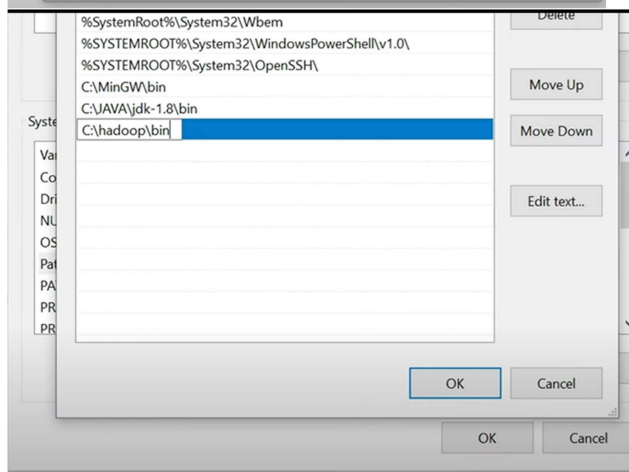
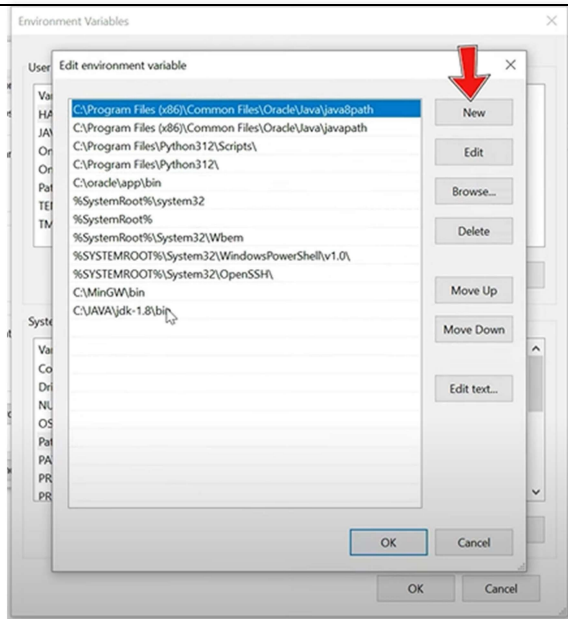
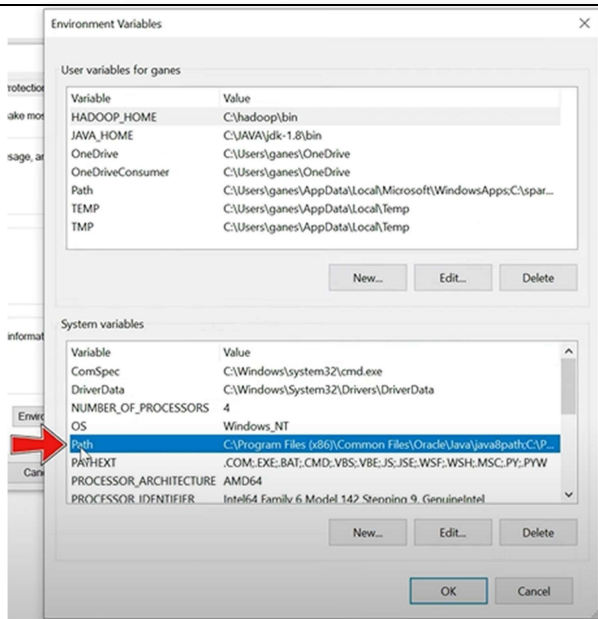


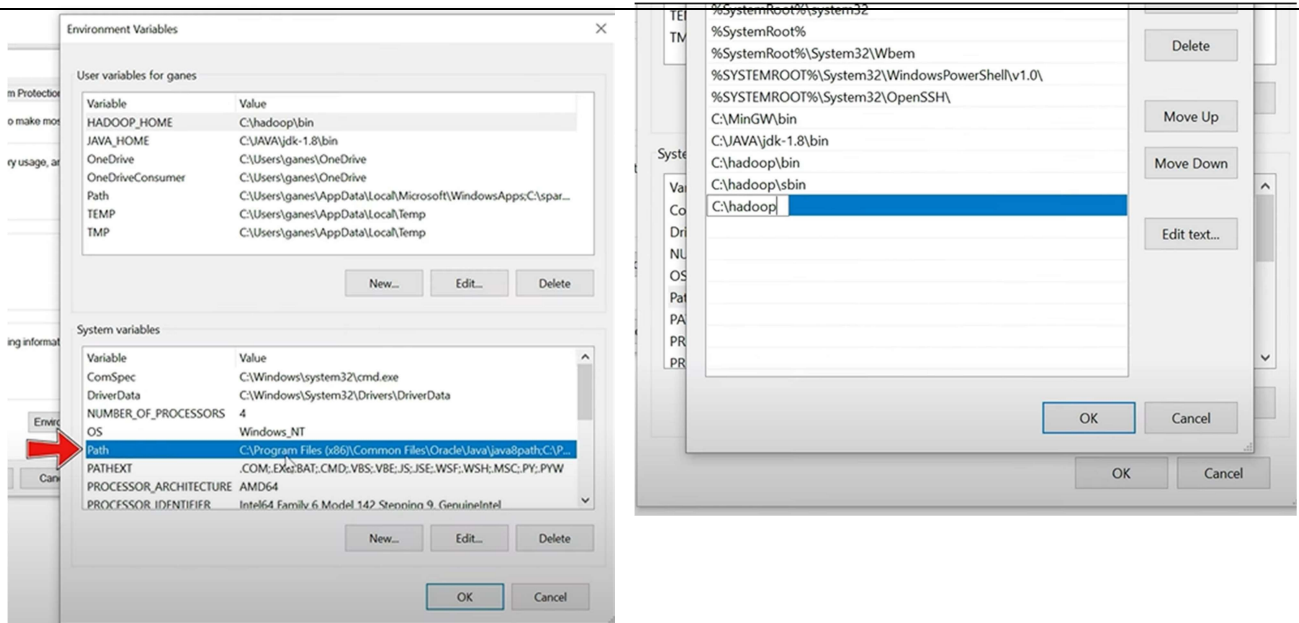


8. Then choose an option “Advanced”, in that , select “Environment Variable”.
9. There are two windows that will be available, i.e “User variables” and “System Variables”.
In “User variables” window, create two new variables with values, i.e
Variable Name: JAVA_HOME
Variable Value: C:\JAVA\jdk-1.8\bin
Variable Name: HADOOP_HOME
Variable Value: C:\hadoop\bin
10. In “System variables” window, choose an option called as “Path” by double clicking on it.
Add the below given paths to it by choosing an option “New”:
C:\JAVA\jdk-1.8\bin
C:\hadoop\bin
C:\hadoop\sbin
C:\hadoop
Configuration of hadoop and Java is done successfully.
Refer the given screen shots below.









Verify whether Java and Hadoop has been installed or not by typing below given commands in Command Prompt, by choosing “run as an administrator”.

java -version
hadoop version

```
C:\Users\Vinayaka>java -version
java version "1.8.0_411"
Java(TM) SE Runtime Environment (build 1.8.0_411-b09)
Java HotSpot(TM) 64-Bit Server VM (build 25.411-b09, mixed mode)
```

```
C:\Users\Vinayaka>hadoop version
Hadoop 3.2.4
Source code repository Unknown -r 7e5d9983b388e372fe640f21f048f2f2ae6e9eba
Compiled by ubuntu on 2022-07-12T11:58Z
Compiled with protoc 2.5.0
From source with checksum ee031c16fe785bbb35252c749418712
This command was run using /C:/hadoop/share/hadoop/common/hadoop-common-3.2.4.jar
```

- Go to “Downloads”, select “Technical Windows” and in that, select “Software files”, extract files in the same location.
- Install the file named “VC_redist.x64”, it is mandatory to install. “Microsoft Visual C++ 2015-2022 Redistributable (x64)”
- Open “Command Prompt” by right clicking on it and choose “Run as Administrator”.
- To format the namenode, type a command as given below:
hdfs namenode -format

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This command is used to format the NameNode in Hadoop's HDFS (Hadoop Distributed File System). It initializes the metadata storage for HDFS.

When to use the command “hdfs namenode –format”?

- When setting up HDFS for the first time.
- When resetting the filesystem (erasing all metadata).
- When the NameNode metadata is corrupted and needs re-initialization.

How command “hdfs namenode –format” works?

- It deletes existing metadata of the NameNode.
- It creates a new namespace ID for HDFS.
- It initializes the NameNode storage directories.

```
C:\Users\Vinayaka>hdfs namenode -format
2025-03-01 18:59:59,442 INFO namenode.NameNode: STARTUP_MSG:
/*****
STARTUP_MSG: Starting NameNode
STARTUP_MSG: host = DESKTOP-PKV2H5J/192.168.0.110
STARTUP_MSG: args = [-format]
STARTUP_MSG: version = 3.2.4
STARTUP_MSG: classpath = C:\hadoop\etc\hadoop;C:\hadoop\share\hadoop\com
animal-sniffer-annotations-1.17.jar;C:\hadoop\share\hadoop\com
common\lib\avro-1.7.7.jar;C:\hadoop\share\hadoop\common\lib\cl
on\lib\commons-cli-1.2.jar;C:\hadoop\share\hadoop\common\lib\c
ommon\lib\commons-compress-1.21.jar;C:\hadoop\share\hadoop\com
```

Execute the below given commands to change the directory to sbin.

1. **cd **
Moves to the root directory (C:\) from the current location. This ensures navigation starts from the base of the file system.
2. **cd hadoop**
Moves into the hadoop directory located in C:\
3. **cd sbin**
Moves into the **sbin** (system binaries) folder inside hadoop.
This is where Hadoop administrative scripts (like `start-dfs.sh`, `stop-dfs.sh`) are located.

```
C:\>cd hadoop
C:\hadoop>cd sbin
C:\hadoop\sbin>
```

- When ever “**start-dfs.cmd**” command gets executed, two windows will be opened, that is “Data node” and “name node” which will be executing its task, wait till execution of task completes.

About start-dfs.cmd

- This command is used to **start the Hadoop Distributed File System (HDFS)** in Windows.
- It should be executed from the **sbin** directory inside the Hadoop installation folder.

What It Does?

- Starts the NameNode – Manages HDFS metadata.
- Starts the DataNodes – Stores actual data.

Adding Files and Directories to HDFS

- 1) Create a directory on hadoop distributed file system:
- 2) Go to browser and check whether the directory got created or not: type “localhost:9870”, after the page opens, click on “utilities” and there choose “Browse the File Systems”.

```
C:\Windows\system32>hadoop fs -mkdir /doc
```

Copy a file from local system to HDFS:

- 1) Create a text file in one of the folder, in any location.
- 2) Type a command:

```
C:\Windows\system32>hadoop fs -put E:\File1\Textdoc.txt /doc
```

Use the path of your file where it is created. And the folder that is created in HDFS.

- 3) To verify Textdoc.txt has been copied to HDFS or not, go to browser, type “localhost:9870” -> “utilities” -> “Browse the File System”. There in the window, a folder have been created by the name “doc”, open that and check for the file named “Textdoc.txt”.

Retrieve a file from HDFS

- 1) Type a command as given:

```
C:\Windows\system32>hadoop fs -ls /doc
Found 1 items
-rw-r--r--  3 Vinayaka supergroup      41 2025-03-02 08:33 /doc/Textdoc.txt
```

which will list out the files in “doc” folder. As there was only 1 file present, so its showing “found 1 items”

- 2) Retrieve a file from HDFS to local system:

```
C:\Windows\system32>hadoop fs -get /doc/Textdoc.txt E:\File2
```

```
C:\Windows\system32>
```

so, here, the file named “Textdoc.txt”

which was present in “doc” directory of hdfs moved to a new path “E:\File2”, where File2 folder should be created in prior before using the same folder name in the – command.

Deleting Files and Directories in HDFS

- 1) Delete a file from HDFS:

```
C:\Windows\system32>hadoop fs -rm /doc/Textdoc.txt
Deleted /doc/Textdoc.txt
```

- 2) Delete a directory (including files inside):

```
C:\Windows\system32>hadoop fs -rm -r /doc
Deleted /doc
```